

Unit # 6

Representing Vectors in Two-Space and Three-Space

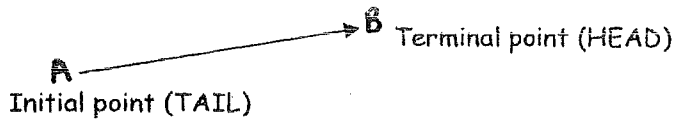
VECTOR CONCEPTS

Vectors are quantities having both magnitude and direction. For example, a car is traveling with a velocity of 30 km/h from the north-west.

Scalars are quantities having magnitude only. Temperature, mass and height are examples as there is no direction associated with these.

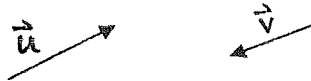
Note: Displacement is a vector quantity (because it refers to traveling a certain distance in a particular direction), whereas distance is a scalar quantity (because there is no direction associated with it).

Representing a vector:



Length and direction should be reflected in the vector diagram. Thus, a vector with twice the magnitude of another vector should be drawn with two times the length.

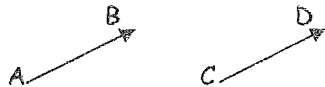
Symbols used:



Note: For length (magnitude), we use absolute value brackets when we have positive scalars. For example,

Vector Equality:

Two vectors are equal if and only if they have the same magnitude and same direction.



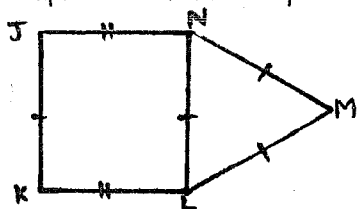
Two vectors are opposite if they have the same magnitude but point in opposite directions.



When two vectors are opposite, such as above, one is the negative of the other:

Two vectors are parallel if their directions are either the same or opposite. They are called collinear vectors.

Example #1: State all equal vectors in the diagram below.



Example #2: Use a suitable scale to draw vectors representing each of the following:

- (a) A wind from the north-west at 25 km/h. (b) A trip of 10 km due east followed by 5 km due south.

To specify the direction of a vector, state the angle it makes with another vector or with some given direction such as the horizontal axis, the vertical axis or a compass direction.

The angle between two vectors is the angle (no larger than 180 degrees) formed when the vectors are placed tail to tail (ie. Starting at the same point).



Scalar Multiplication:

This is used for 2 vectors that are parallel (where k is a scalar (ie. A real number)).



M is the midpoint of the line segment AB .

Vector equations:

Unit Vectors:

These are vectors with magnitude one (denoted by a carat (^) placed over the symbol).

1. A unit vector in the direction of any vector can be found by dividing the vector by its magnitude: $\hat{v} = \frac{v}{|v|}$
2. Any vector can be expressed as the product of its magnitude and a unit vector in the direction of the vector: $\vec{v} = |v|\hat{v}$

Zero vector:

The zero vector has a magnitude of 0 (direction is undefined). It is denoted as $\vec{0}$ and is represented by a point.

VECTOR LAWS

In many applications of vectors, we need to find the combined effect or sum of 2 or more vectors. For example, the effect that wind velocity has on the velocity of an aircraft.

Triangle Law of Vector Addition:

For vectors $\vec{u}$ and $\vec{v}$, the sum (resultant) is a vector from the tail of $\vec{u}$ to the tip of $\vec{v}$, when the tail of $\vec{v}$ is placed at the tip of $\vec{u}$:

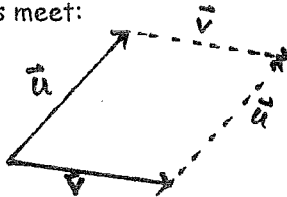


Note: the sum of $\vec{u}$ and $\vec{v}$ is $\vec{u} + \vec{v}$

The order in which we add is not important. Thus, vector addition is commutative: $\vec{u} + \vec{v} = \vec{v} + \vec{u}$

Parallelogram Law of Vector Addition:

The sum $\vec{u} + \vec{v}$ is the diagonal of the parallelogram from the point where the tails are joined to the point where their heads meet:



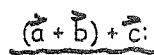
Example #1: Jamie travelled 3 km north then turned to travel 2.5 km due east. Keri leaves after Jamie and decides to take the most direct route to meet Jamie.

(a) Draw a diagram of this situation.

(b) Describe Keri's path in terms of length and direction.

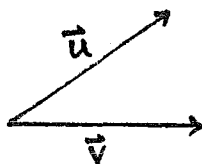
Associative Law of Addition:

$\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$ This means that we can omit the brackets and write them both equal to $\vec{a} + \vec{b} + \vec{c}$



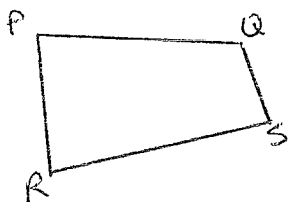
Vector Subtraction:

For vectors $\vec{u}$ and $\vec{v}$, the difference, $\vec{u} - \vec{v}$, is $\vec{u} + (-\vec{v})$. That is, the difference is the same as adding the opposite.



Note: The sum of a vector and its opposite is a vector with zero magnitude. That is, it is a zero vector as $\vec{u} + (-\vec{u}) = \vec{0}$.

Example #2: Given the diagram below, show that $\vec{PQ} - \vec{RQ} = \vec{PR}$.

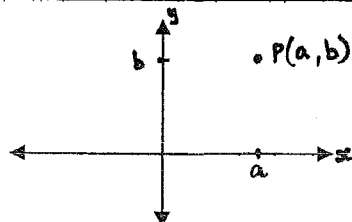


Example #3: Find the magnitude and direction of the sum of two vectors $\vec{u}$ and $\vec{v}$ if their magnitudes are 5 and 8 units respectively, and the angle between them is 30 degrees.

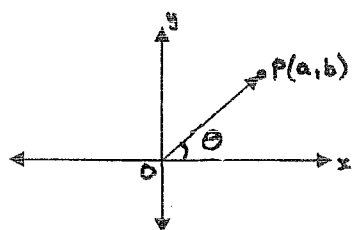
ALGEBRAIC VECTORS

Algebraic vectors allow the use of algebraic methods in the study of vectors.

Vectors in 2-dimensions (in a plane):



Point P in the plane is determined by the ordered pair which are its coordinates.



$\overrightarrow{OP}$ is the position vector of a point P from the origin to the point.

$$\vec{u} = \overrightarrow{OP} = [a, b]$$

The ordered pair (a, b) is called an algebraic vector. The values of "a" and "b" are the "x" and "y" components of the vector.

The magnitude is $|\vec{u}| = \sqrt{a^2 + b^2}$.

The direction is $\theta = \tan^{-1}\left(\frac{b}{a}\right)$ with $0^\circ \leq \theta \leq 360^\circ$ measured counter-clockwise from the positive x-axis to the line of the vector. The actual value depends on the quadrant in which $P(a, b)$ lies.

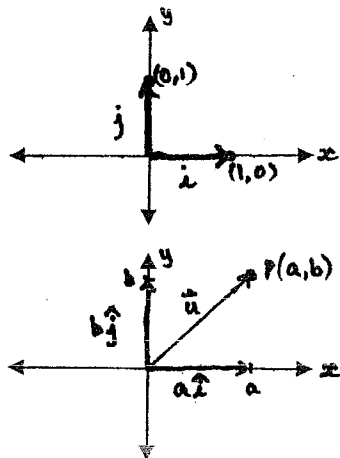
Example #1: Given point $P(-2, -5)$,

(a) draw the position vector

(b) express the position vector in ordered pair notation

(c) determine the magnitude and direction

Notation using unit vectors:



$\hat{i}$ is the unit vector along the positive x-axis and $\hat{j}$ is the unit vector along the positive y-axis.

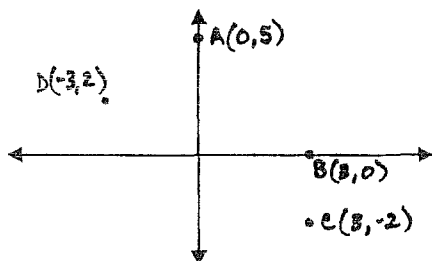
Note: $\hat{i} = [1, 0]$ and $\hat{j} = [0, 1]$

Ordered pair notation and unit notation are equivalent.

$$\vec{u} = \overrightarrow{OP} = [a, b] \text{ or } \vec{u} = \overrightarrow{OP} = a\hat{i} + b\hat{j} \quad \text{thus, } [a, b] = a\hat{i} + b\hat{j}$$

Vectors $a\hat{i}$ and $b\hat{j}$ are *vector components* of $\overrightarrow{OP}$ along $\hat{i}$ and $\hat{j}$ respectively.
a and b are the *scalar components* of $\overrightarrow{OP}$.

Example #2: Express the position of each of the points shown as an ordered pair and in unit vector notation.



Vector Addition and Subtraction in Component form:

$$\text{If } \vec{u} = [u_1, u_2] \text{ and } \vec{v} = [v_1, v_2] \text{ then } \vec{u} + \vec{v} = [u_1 + v_1, u_2 + v_2] \text{ and } \vec{u} - \vec{v} = [u_1 - v_1, u_2 - v_2]$$

Scalar Multiplication in Component form:

$$\text{If } \vec{u} = [u_1, u_2] \text{ and } k \in \mathbb{R}, \text{ then } k\vec{u} = [ku_1, ku_2]$$

Two algebraic vectors are equal if and only if their respective Cartesian components are equal.

Example #3: If $\vec{u} = [2, -3]$ and $\vec{v} = [5, -3]$, find (a) $\vec{u} + \vec{v}$ (b) $-3\vec{u}$

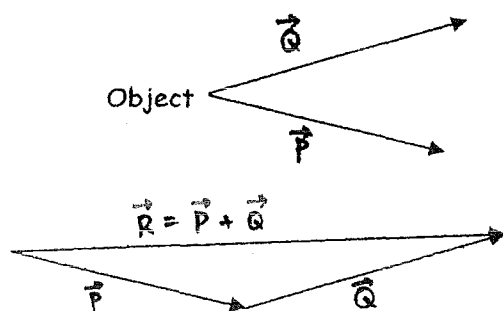
Example #4: If $P_1 = (2, -3)$ and $P_2 = (-5, -2)$, find $\overrightarrow{P_1P_2}$

Example #5: Using vectors, show that the three points A(5, -1), B(-3, 4) and C(13, -6) are collinear.

FORCE AS A VECTOR

Force is a push or pull on an object. Forces are measured in Newtons (N). Force due to gravity on a mass of 1kg is about 9.8N. That is, a mass of 1kg is said to weigh 9.8N.

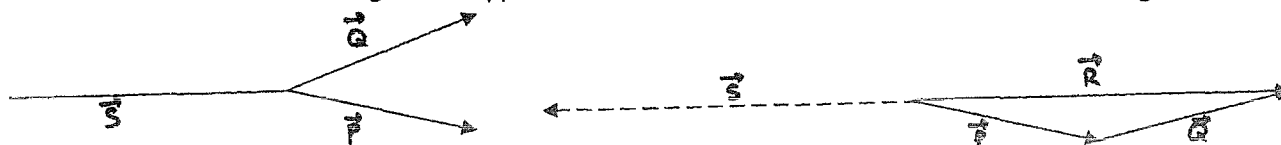
Forces can be combined in the same way that vectors are added:



$\vec{R}$ is called the resultant of $\vec{P}$ and $\vec{Q}$. That is, the resultant of the number of forces is the vector sum of these forces.

If an object does not change its motion (or its state of rest), then the object remains in a state of equilibrium. That is, it must be acted upon simultaneously by a force that counteracts or cancels the effect of $\vec{R}$.

Consider the force $\vec{S}$ acting in the opposite direction to that of $\vec{R}$ but with the same magnitude:



The particle/object is in equilibrium under the effect of $\vec{P}$, $\vec{Q}$ and $\vec{S}$ or under the effect of $\vec{R}$ and $\vec{S}$. $\vec{S}$ is called the equilibrant of $\vec{R}$. Conversely, $\vec{R}$ is called the equilibrant of $\vec{S}$.

Note: Whenever a particle is in equilibrium, the vector sum of all forces acting on it is $\vec{0}$.

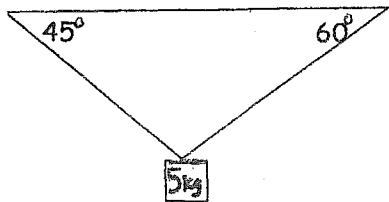
Example #1: A force of 20N acts west and a force of 10N acts north on a particle. Find the direction and magnitude of the resultant and the equilibrant of these 2 forces.

Example #2: Two draft horses pull a load. The chain between the horses and the load are at an angle of 60 degrees to each other. One horse pulls with a force of 230N and the other with a force of 340N.

(a) What is the resultant force on the load? In what direction is this force?

(b) What is the equilibrant force on the load? In what direction is this force?

Example #3: A traffic sign with a mass of 5kg is suspended above the street as shown below. Find the tensions (forces) in each of the wires.



VELOCITY AS A VECTOR

Speed is the rate of change of the distance with respect to time. It is a *scalar quantity*.

Velocity is the distance travelled per unit of time where the direction and magnitude (speed) are important. It is a *vector quantity*.

Example #1: The air speed of a small plane is 200 km/h steering $N40^\circ E$. The wind speed is 50 km/h from the west. Find the ground speed and course of the plane.

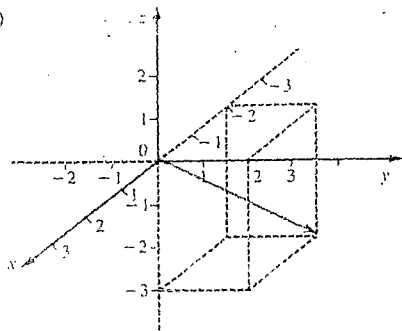
Example #2: A canoeist wishes to cross a river 1 km in width. If the current flows at 5 km/h and the canoeist can paddle at 13 km/h in still water, determine the:

- (a) direction the canoeist must head to travel straight across the river.
- (b) time it will take to cross the river.

Review Vectors

5. If $\vec{u} = (-2, 3, 1)$ and $\vec{v} = 2\vec{i} - 3\vec{j} + 4\vec{k}$ find
 - (a) $3\vec{u} - 4\vec{v}$
 - (b) $|-2\vec{u} - \vec{v}|$
6. Given the points $A(-2, 3, 3)$, $B(-4, -2, 3)$, and $C(3, -1, 4)$, find the perimeter of $\triangle ABC$.
7. Find two unit vectors collinear with $(-2, 3, -3)$.
8. Use the definition of dot product, $\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$, to show
 - (a) $\vec{j} \cdot \vec{j} = 1$
 - (b) $\vec{i} \cdot \vec{j} = 0$
 - (c) $\vec{k} \cdot \vec{i} = 0$
9. Find the dot product of the following pairs of vectors.
 - (a) $\vec{u} = (2, -5)$ $\vec{v} = (-4, -9)$
 - (b) $\vec{u} = -4\vec{i} + 12\vec{j}$ $\vec{v} = 5\vec{i} - 6\vec{j}$
 - (c) $\vec{u} = (-1, -2, 3)$ $\vec{v} = (-6, 5, -5)$
 - (d) $\vec{u} = 5\vec{i} + 3\vec{j} + 7\vec{k}$ $\vec{v} = 8\vec{i} - 7\vec{k}$
10. Find the angle θ between the two given vectors. Give your answer to the nearest degree.
 - (a) $\vec{u} = (-3, 6)$ $\vec{v} = (6, -12)$
 - (b) $\vec{u} = 7\vec{i} - 10\vec{j}$ $\vec{v} = 8\vec{i} + 2\vec{j}$
 - (c) $\vec{u} = (-1, 2, -3)$ $\vec{v} = (-5, 10, -15)$
 - (d) $\vec{u} = -3\vec{i} - 4\vec{j} - 6\vec{k}$ $\vec{v} = -4\vec{j} + 9\vec{k}$
11. If $\vec{u} = (-3, 6, a)$ and $\vec{v} = (b, -2, 4)$ are orthogonal, find a relationship between a and b .
12. Use the definition of the cross product, $\vec{u} \times \vec{v} = |\vec{u}| |\vec{v}| \sin \theta \hat{n}$, to show
 - (a) $\vec{i} \times \vec{i} = \vec{0}$
 - (b) $\vec{k} \times \vec{j} = -\vec{i}$
 - (c) $\vec{j} \times \vec{k} = \vec{i}$
13. For each of the following pairs of vectors find $\vec{u} \times \vec{v}$.
 - (a) $\vec{u} = (2, 7, -1)$ $\vec{v} = (-4, 5, 1)$
 - (b) $\vec{u} = (1, 1, 1)$ $\vec{v} = (-2, -2, -2)$
 - (c) $\vec{u} = 3\vec{i} - \vec{j} + 2\vec{k}$ $\vec{v} = -\vec{i} + \vec{j} + 3\vec{k}$
14. Find a unit vector perpendicular to both $\vec{u} = (1, -1, 1)$ and $\vec{v} = (1, 1, 1)$. Check your answer using the dot product.
15. The points $P(1, 1, 1)$, $Q(3, -3, 4)$, and $S(-2, 4, 5)$ are three vertices of parallelogram $PQRS$.
 - (a) Find the coordinates of R .
 - (b) Find the measures of the interior angles.
 - (c) Find the area of the parallelogram.
16. If $\vec{u} = (1, -3, 4)$, $\vec{v} = (-1, -2, 2)$, and $\vec{w} = (2, 1, -1)$, find
 - (a) $(\vec{u} \times \vec{v}) \cdot \vec{w}$
 - (b) $(\vec{u} \times \vec{v}) \times \vec{w}$
 - (c) What conclusion can be made concerning the vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$?
17. If $\vec{u} = (1, -1, 2)$, $\vec{v} = (2, -1, -1)$, and $\vec{w} = (1, -2, 6)$, find
 - (a) $(\vec{u} \times \vec{v}) \cdot \vec{w}$

1. (c)



2. (a) collinear vectors lying on the z -axis
(b) coplanar vectors lying in the xz -plane
3. $c = \frac{15}{2}$ and $b = -\frac{5}{2}a$
4. (a) $(0, -13, 26)$ (b) $-3\vec{i} - 7\vec{j} + 10\vec{k}$ or $(-3, -7, 10)$ (c) $\sqrt{29}$ (d) $2\sqrt{85}$
5. (a) $(-14, 21, -13)$ (b) 7
6. $\sqrt{29} + \sqrt{51} + \sqrt{42}$
7. $\left(-\frac{2}{\sqrt{22}}, \frac{3}{\sqrt{22}}, -\frac{3}{\sqrt{22}}\right)$ or $\left(\frac{2}{\sqrt{22}}, -\frac{3}{\sqrt{22}}, \frac{3}{\sqrt{22}}\right)$
9. (a) 37 (b) -92 (c) -19 (d) -9
10. (a) 180° (b) 69° (c) 0° (d) 120°
11. $4a - 3b = 12$
13. (a) $(12, 2, 28)$ (b) $(0, 0, 0)$ (c) $(-5, -11, 2)$
14. $\left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$ or $\left(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right)$
15. (a) $(0, 0, 8)$ (b) 79° and 101° (c) $5\sqrt{38}$
16. (a) 3 (b) $(11, -8, 14)$ (c) They are not coplanar.
17. (a) -1 (b) $(32, -17, -11)$ (c) They are not coplanar.
18. (a) coplanar (b) non-coplanar
(c) non-coplanar
19. (a) not a basis (b) not a basis (c) basis

CHAPTER 5 EQUATIONS OF LINES

REVIEW AND PREVIEW TO CHAPTER 5

EXERCISE 1

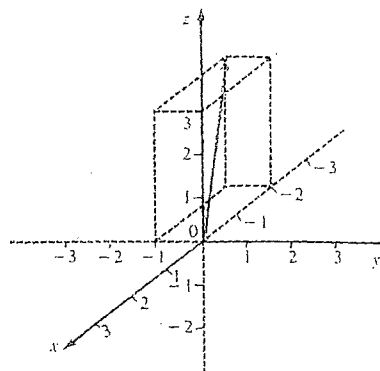
1. (a) $y = -2x + 13$ (b) $y = 5$
(c) $y = 3x - 7$
2. (a) $5x - 2y - 7 = 0$
(b) $4x - 9y - 38 = 0$
(c) $12x + 5y - 1 = 0$ (d) $x + 2y = 0$

19. (a) $\vec{u}$ and $\vec{v}$ are collinear, $\vec{v}$, $\vec{w}$, and $\vec{x}$ are coplanar. (b) No two vectors are collinear, $\vec{u}$, $\vec{w}$, and $\vec{x}$ are coplanar and $\vec{v}$ does not lie in this plane. (c) No two vectors are collinear. No three vectors are coplanar.

21. (a) $\vec{u}$, $\vec{v}$, and $\vec{x}$ form a basis and $\vec{w} = 2\vec{u} + \vec{v} + 0\vec{x}$.
(b) $\vec{u}$, $\vec{v}$, and $\vec{w}$ form a basis and $\vec{x} = 0\vec{u} - 2\vec{v} + 0\vec{w}$.
(c) $\vec{u}$, $\vec{v}$, and $\vec{w}$ form a basis and $\vec{x} = \frac{1}{2}\vec{u} + \frac{1}{2}\vec{v} + 0\vec{w}$.

4.7 CHAPTER 4 TEST

1.



2. (a) $(-10, -3, 5)$ (b) $\sqrt{91}$
(c) $\left(\frac{2}{\sqrt{14}}, -\frac{3}{\sqrt{14}}, -\frac{1}{\sqrt{14}}\right)$ (d) 4
(e) $(6, 0, 12)$
3. (a) $(0, -2, 1)$ (b) 51° and 129° (c) $\sqrt{185}$
4. $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$ or $\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right)$
6. (a) -38 (b) $(21, -22, 19)$
7. No two vectors are collinear. $\vec{u}$, $\vec{v}$, and $\vec{w}$ are coplanar and $\vec{x}$ does not lie in this plane.
8. $\vec{x} = 2\vec{u} - \vec{v} + 0\vec{w}$, $\vec{x}$ is coplanar with $\vec{u}$ and $\vec{v}$.

3. (a) undefined (b) because slope is undefined
(c) $x = k$

EXERCISE 2

1. (a) $(-1, 2)$ (b) $\left(\frac{5}{2}, -5\right)$ (c) $\left(\frac{1}{2}, \frac{1}{3}\right)$
(d) $(-5, -1)$